

Analysis of Factors Affecting Restaurant Inspection Scores in New York City

Given our focus on 2024, we will limit the dataset to inspections conducted within this year. Some restaurants may have undergone multiple inspections within 2024, so we will adopt strategies to handle these cases effectively.

Research Questions

1. How do inspection scores vary across boroughs and cuisine types?
2. Is there a temporal trend in restaurant inspection scores in New York City?
3. What are the main factors associated with higher inspection scores, indicating poorer health compliance?

Data Description

The dataset for this study, "DOHMH New York City Restaurant Inspection Results," https://data.cityofnewyork.us/Health/DOHMH-New-York-City-Restaurant-Inspection-Results/43n-n-pn8j/about_data includes information on restaurant inspections conducted by the DOHMH. Key variables in the dataset include:

- **score**: The total score assigned during an inspection, with higher values indicating poorer outcomes.
- **boro**: The borough where the restaurant is located (e.g., Manhattan, Brooklyn).
- **cuisine_description**: The type of cuisine offered by the restaurant (e.g., Italian, Chinese).
- **inspection_date**: The date of the inspection.
- **violation_code** and **violation_description**: Codes and descriptions of specific violations recorded during inspections.
- **critical_flag**: A binary indicator showing whether a violation is considered critical, which may be linked to higher scores.
- **inspection_type**: The type of inspection performed (e.g., initial, follow-up).

4. Methodology

Question 1

- **Average Score Calculation**: For restaurants with multiple inspections in 2024, we will calculate the average score across all inspections within the year.
- **Borough and Cuisine Analysis**: Calculate the average inspection score by borough and cuisine type to identify any spatial or cultural patterns in health compliance.
- **Visualization**: Use bar charts or box plots to visualize the average inspection scores by borough and cuisine type. These visualizations will highlight variations in compliance across different areas and food types.

Question 2

- **Time Series Analysis**: Conduct a simple **time series analysis** on average scores by month to determine if there are significant fluctuations in inspection scores within 2024, which may point to seasonal trends or temporal variations.

Question 3

- **Multiple Regression Analysis:** Perform a multiple regression analysis to identify factors associated with higher inspection scores. The dependent variable will be score, while independent variables will include (For this analysis, we will retain only the **most recent inspection in 2024** for each restaurant, as the latest score is most indicative of the restaurant's current compliance status.)